Poster Number 25W0900

Converting light emulsion into a creamy foam for psoriasis treatment. A case study with betamethasone dipropionate.

E. Dauphin-Chanard; C. André ; D. Marchaud
GATTEFOSSÉ - Saint-Priest – France - edauphin@gattefosse.com

2016 AAPS ANNUAL MEETING AND EXPOSITION
NOVEMBER 13–17, 2016
COLORADO CONVENTION CENTER, DENVER
aaps

PURPOSE

Psoriasis is a skin inflammation disorder mainly characterized by redness, inflamed lesions and scaly plaques[1]. According to the WHO, this 'non communicable, painful, disfiguring and disabling disease' affects approximately 100 million people worldwide[2]. Besides physical symptoms, psoriasis is associated with psychosocial problems and comorbidities that can impair quality of life for patients[2,3]. In this context, adherence to the treatment is the primary objective to achieve[4,5].

Foams appear to be appropriate vehicles to ensure patient compliance and hence therapy effectiveness, thanks to their sensorial attributes: easy spreadability, easy application on sensitive or highly inflamed skin, rapid absorption and penetration and attractive texture[6].

This study aims to develop emollient foam containing betamethasone dipropionate (highly potent corticosteroid) and exhibiting pleasant sensorial characteristics.

METHOD

Materials

Glycerol monostearate and PEG-75 stearate (Gelot™ 64), PEG-6 stearate (and) glycol stearate (and) PEG-32 stearate (Tefose® 63), Triceteareth-4 phosphate (and) ethylene glycol stearates (and) diethylene glycol stearates (Sedefos™ 75), as O/W emulsifiers; oleoyl polyoxylglycerides (Labrafil® M 1944 CS), lauroyl macrogol-6 glycerides (Labrafil® M 2130 CS), medium chain triglycerides (Labrafac™ Lipophile WL 1349), as oily phase; diethylene glycol monoethyl ether (Transcutol® P), as solubilizer; are provided by Gattefossé. Propylene glycol, as humectant, is supplied by Univar. Ethanol, Isopropanol, isopropyle myristate, oleic acid, as solvents; benzyl alcohol, as preservative; and sodium hydroxide, as buffer media; are purchased at Sigma-Aldrich. Betamethasone dipropionate, used as model drug, is sold by Kejiang Xianju Pharmaceutical Co.

Methods

The solubility of BP is assessed at 25°C after two weeks in each individual excipient. Drug concentration is determined by HPLC (Waters chromatograph Alliance 2695D, coupled with a 2487 model UV detector at 254 nm).
The emulsions are characterized at least 24 hours after their production to allow maturation time. pH (Mettler Toledo MP225 pH meter) and viscosity at 25°C (TVe-05 viscosimeter, Lamy) are measured. Microscopic observations are carried out to determine the emulsion's homogeneity and droplet size (Zeiss AX10 Scope A1 microscope).
Light emulsions are placed into pressurized aerosols to evaluate their foaming capacity. Quality and persistency of the foams were visually assessed instantly and after 3 minutes (t_0 and t_3).

CONCLUSION

This study highlights the benefits of using Tefose® 63: it enables the formation of a stable and light emulsion that generates elegant foam using a pressurized aerosol. Simple formulation and high sensorial quality foams are obtained with betamethasone dipropionate.

Moreover, foams bring advantages compared to the traditional vehicles for treatment of topical disorders: preferred sensorial attributes, safety and efficiency[7]. Foams appear to be a promising and attractive vehicle for the long-term management of psoriasis. These vehicles gather patient expectation and skin disease requirements: easy spreadability, easy application on sensitive or highly inflamed skin, rapid absorption and penetration, delivery of constant volume and attractive texture.

RESULTS AND DISCUSSION

The solubility study underlines the powerful solubilizing properties of Transcutol® P: 2% (W/W) are sufficient to fully solubilized betamethasone dipropionate.

Among all emulsifiers tested, Tefose® 63 provides better emulsifying properties and excellent foaming capacity. The evaluation of the different oily vehicles emphasizes the benefits of using Labrafil® M 1944 CS (3% W/W) as a co-emulsifier. At a ratio 2/1 (emulsifier/ co-emulsifier), the stability of the emulsion is improved. As humectant, propylene glycol enhances the foam's sensorial attributes. Sodium hydroxide is required to achieve a pH around 8.

The optimized formulation is composed of: 0.06% betamethasone dipropionate, 6.0% Tefose® 63, 3.0% Labrafil® M 1944 CS, 5.0% Propylene glycol, 2.0% Transcutol® P, 1.0% benzyl alcohol, 0.05% sodium hydroxide (sol. 10%) and 82.89% demineralized water.

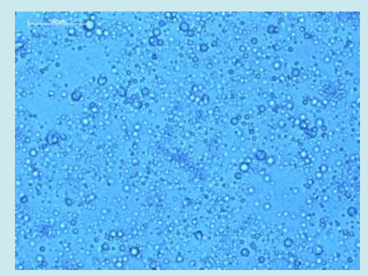
The white emulsion is homogenous and light. This low viscosity enables the production of a creamy and persistent foam using a pressurized aerosol (Table 1).
The microscopic assessment shows small droplets (1.8 ± 1.2 µm) and confirms that betamethasone dipropionate is fully solubilized. The process has no impact on betamethasone dipropionate as no crystals are microscopically visible in the foam.

Compared to conventional dosage forms, betamethasone dipropionate foam exhibits better sensorial features: light texture, easy spreadability, easy application, minimum of rub-in, convenient for large skin area, suitable for sensitive, highly inflamed skin or hair bearing skin, delivery of constant volume, rapid absorption and penetration.

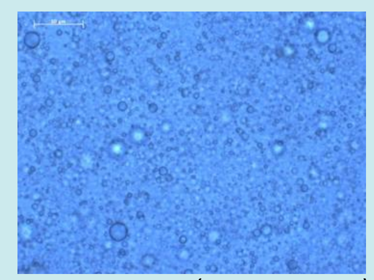
The foam formulated with Tefose® 63 brings a higher comfort during application with a cosmetic and attractive texture. These emollient characteristics can address patient compliance for efficient treatment.

Composition	Characterization
0.06% Betamethasone dipropionate 6.00% Tefose® 63 3.00% Labrafil® M 1944 CS 5.00% Propylene glycol 2.00% Transcutol® P 1.00% Benzyl alcohol 0.05% Sodium hydroxide (sol. 10%) 82.89% Demineralized water	pH 8.0 Viscosity (25°C) = 460 mPa.s Droplet size = 1.8 ± 1.2 µm

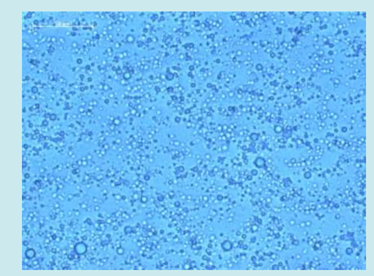
Microscopy



t0 *(Magn. X50)*

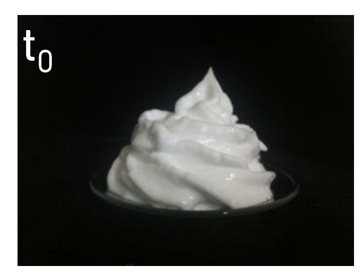


t1month *(Magn. X40)*



t3months *(Magn. X50)*

Macroscopy



REFERENCES

1. http://www.drugs.com/condition/psoriasis.html. Accessed on 2016, June 10.
2. Global report on psoriasis - World Health Organization 2016.
3. https://www.psoriasis.org/about-psoriasis/causes. Accessed on 2016, June 13.
4. Lebwohl M. & al, Fixed Combination Aerosol Foam Calcipotriene 0.005% (Cal) Plus Betamethasone Dipropionate 0.064% (BD) is More Efficacious than Cal or BD Aerosol Foam Alone for Psoriasis Vulgaris - A Randomized, Double-blind, Multicenter, Three-arm, Phase 2 Study - J Clin Aesthet Dermatol. 2016;9(2):34–41.
5. Bewley A, Page B. Maximizing patient adherence for optimal outcomes in psoriasis. J Eur Acad Dermatol Venereol. 2011; 25 (suppl 4):9–14.
6. Arzhavitina A. and Steckel H. Foams for pharmaceutical and cosmetic application. Int. J. Pharm.; 394: 1-17 (2010)
7. Advances in Psoriasis Treatment.WebMD Feature. Reviewed by Stephanie S. Gardner, MD. Reviewed on 2015, November 15. Accessed on 2016, June 14.

GATTEFOSSÉ